SYNTOMIC RING MAPS ARE EXACTLY THE FLAT LOCAL COMPLETE INTERSECTIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Characterisations of syntomic maps and the reduction to a global complete intersection over a field. Algebra, Lemma \ref{algebra-lemma-syntomic}, Algebra, Definition \ref{algebra-definition-lci}, and Algebra, Definition \ref{algebra-definition-lci-field}, cited at proof lines 8417-8418 and 8427. Reduce the verification to the case of a base change to a field.
- A relative global complete intersection is a local complete intersection. Lemma \ref{lemma-relative-global-complete-intersection-koszul} and Lemma \ref{lemma-lci-local}, cited at proof lines 8417-8420. Gives one direction of the equivalence.
- Flat base change turns a Koszul regular sequence into an H1-regular sequence generating the base changed ideal. Lemma \ref{lemma-relative-regular-immersion-algebra}, cited at proof lines 8441-8442. Transports the local presentation of the ideal through the base change to the residue field.
- A finite module cut out by an H1-regular sequence at a prime is cut out by a regular sequence there. Lemma \ref{lemma-noetherian-finite-all-equivalent} and Algebra, Lemma \ref{algebra-lemma-lci}, cited at proof lines 8445-8448. Upgrade the base changed presentation to a global complete intersection over the field.

Source reference. Original lemma 07D3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9398–9398. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00SL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1043–1043. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00SY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1056–1056. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07D1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9396–9396. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07D2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9397–9397. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00S9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1031–1031. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07CV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9390–9390. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0CEP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15584–15584. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 09CC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11825–11825. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00SC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1034–1034. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $R \to S$ be a ring map. The following are equivalent*

(1) *$R \to S$ is syntomic (Algebra, Definition 00SL), and*
(2) *$R \to S$ is flat and a local complete intersection.*

Proof. Assume (1). Then $R \to S$ is flat by definition. By Algebra, Lemma 00SY and Lemma 07D1 we see that it suffices to show a relative global complete intersection is a local complete intersection homomorphism which is Lemma 07D2.

Assume (2). A local complete intersection is of finite presentation because a Koszul-regular ideal is finitely generated. Let $R \to k$ be a map to a field. It suffices to show that $S' = S \otimes_R k$ is a local complete intersection over k, see Algebra, Definition 00S9. Choose a prime $\mathfrak{q}' \subset S'$. Write $S = R[x_1, \ldots, x_n]/I$. Then $S' = k[x_1, \ldots, x_n]/I'$ where $I' \subset k[x_1, \ldots, x_n]$ is the image of I. Let $\mathfrak{p}' \subset k[x_1, \ldots, x_n]$, $\mathfrak{q} \subset S$, and $\mathfrak{p} \subset R[x_1, \ldots, x_n]$ be the corresponding primes. By Definition 07CV exists an

$g \in R[x_1, \ldots, x_n]$, $g \notin \mathfrak{p}$ and $f_1, \ldots, f_r \in R[x_1, \ldots, x_n]_g$ which form a Koszul-regular sequence generating I_g. Since S and hence S_g is flat over R we see that the images $f'_1, \ldots, f'_r$ in $k[x_1, \ldots, x_n]_g$ form a H_1-regular sequence generating I'_g, see Lemma 0CEP. Thus $f'_1, \ldots, f'_r$ map to a regular sequence in $k[x_1, \ldots, x_n]_{\mathfrak{p}'}$ generating $I'_{\mathfrak{p}'}$ by Lemma 09CC. Applying Algebra, Lemma 00SC we conclude $S'_{gg'}$ for some $g' \in S$, $g' \notin \mathfrak{q}'$ is a global complete intersection over k as desired. □